📋 Internship Project — Week 3 & Week 4

Project Title: End-to-End Sales Forecasting & Demand Intelligence System
Assigned Date: [03/07/2026] **Submission Date:** [13/07/2026]

🎯 Problem Statement

Every retail and e-commerce company — from Walmart and Amazon to D-Mart and Flipkart — lives and dies by one question: *"How much of each product will we sell next month, and will we have enough stock to meet that demand?"* Getting this wrong in either direction costs crores — overstock wastes storage and capital, understock loses sales and customers.

This is not a beginner classification problem. This is a **multi-layered, real industry problem** that requires you to work with time-series data (data ordered by date, not rows of independent customers), build and compare multiple forecasting models, detect anomalies in sales patterns, segment products by demand behavior, and deliver a working interactive dashboard that a business manager could open on Monday morning and make stocking decisions from.

You will touch **Time Series Analysis, Machine Learning, Forecasting, Anomaly Detection, Customer/Product Segmentation, and Deployment** — all in one project. This is the kind of system that data science teams at mid-to-large companies actually build and maintain.

Your task: Build an intelligent sales forecasting system that predicts future product demand, detects unusual sales spikes or drops, segments products by demand pattern, and presents everything through a deployed interactive dashboard.

📦 Dataset — Where to Get It

Go to this link and download the dataset:
🔗 https://www.kaggle.com/datasets/rohitsahoo/sales-forecasting

Steps to download:

- Create a free account on www.kaggle.com (if you don't have one)
- Search for **"Superstore Sales Dataset"**
- Download the file — it will be a .csv file named **train.csv**
- Use the full dataset — it covers 4 years of daily sales data across multiple product categories and regions

Supplementary Dataset (Required for Anomaly Detection Task):
🔗 https://www.kaggle.com/datasets/gregorut/videogamesales
(You will use this as a secondary dataset to practice merging and multi-source analysis — a real-world skill since no company keeps all its data in one file)

✅ **Tasks to Complete**
Complete all the following tasks in a Jupyter Notebook (.ipynb):

Task 1 — Data Loading, Merging & Deep Exploration

- Load the Superstore Sales CSV using Pandas
- Parse the **Order Date** and **Ship Date** columns as proper datetime objects
- Extract time features: Year, Month, Week Number, Day of Week, Quarter, Season
- Check for missing values, duplicates, and data type issues
- Aggregate daily sales into **weekly and monthly totals** (you will need both granularities for different models)
- Answer these questions in your notebook with data to back each one:
    - Which product category generates the highest total revenue?
    - Which region has the most consistent sales growth over 4 years?
    - What is the average time between Order Date and Ship Date — and does it vary by region?
    - Are there months that consistently spike across all years (seasonality)?

Task 2 — Time Series Analysis & Decomposition

- Plot the overall monthly sales trend across all 4 years
- Apply **Time Series Decomposition** (using `statsmodels`) to break the sales signal into:
    - Trend component
    - Seasonal component
    - Residual/noise component
- Plot all 4 components clearly on one figure
- Write 3–4 observations: what does the trend tell you? Is seasonality strong or weak? What months show the highest residual noise?
- Check for **stationarity** using the **Augmented Dickey-Fuller (ADF) Test** — explain in plain English what stationarity means and what your test result tells you
- Apply differencing if the series is non-stationary and re-test

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Task 3 — Sales Forecasting using 3 Different Models

This is the core technical task. Build, train, and compare 3 fundamentally different forecasting approaches:

Model 1 — SARIMA (Statistical Model)

- Install statsmodels (`pip install statsmodels`)
- Fit a SARIMA model on monthly sales

- Choose appropriate (p, d, q) and seasonal (P, D, Q, m) parameters — document why you chose them
- Generate a 3-month future forecast with confidence intervals
- Plot actual vs forecasted sales

Model 2 — Facebook Prophet (Industry-standard Forecasting Tool)

- Install Prophet (`pip install prophet`)
- Prepare data in Prophet's required format (ds, y columns)
- Fit the model and generate a 3-month forecast
- Plot the forecast with Prophet's built-in trend and seasonality breakdown
- Extract and interpret the weekly and yearly seasonality components

Model 3 — XGBoost for Time Series (ML-based Approach)

- Convert the time series into a supervised ML problem using lag features:
    - Lag 1 (sales from 1 month ago)
    - Lag 2 (sales from 2 months ago)
    - Lag 3 (sales from 3 months ago)
    - Rolling mean (3-month moving average)
    - Month, Quarter, Season as features
- Train XGBoost Regressor on these features
- Predict the next 3 months
- Plot actual vs predicted

Model Comparison Table (Required)
Create a clear comparison table in your notebook:

Model	MAE	RMSE	MAPE	Forecast for Month 1	Forecast for Month 2	Forecast for Month 3
SARIMA						
Prophet						
XGBoost						

State clearly which model you would recommend for production use and why — based on numbers, not preference.

Task 4 — Product Category & Region Level Forecasting

- Repeat the best performing model (from Task 3) separately for each of the following segments:
    - Furniture category sales

    - Technology category sales
    - Office Supplies category sales
    - West region sales
    - East region sales
- Plot all 5 forecasts together on one comparison chart
- Write: which category/region is showing the strongest upcoming growth according to your model?

Task 5 — Anomaly Detection in Sales Data

- Use **Isolation Forest** (from scikit-learn) to detect anomalous sales weeks — weeks where sales were unusually high or unusually low compared to the expected pattern
- Mark the anomalies on a time series plot (use a different color/marker for anomaly points)
- For each detected anomaly, write a possible real-world explanation (e.g., "spike in November likely corresponds to a festive sale period")
- Apply a second anomaly detection method: **Z-Score based detection** — flag any week where sales deviate more than 2 standard deviations from the rolling mean
- Compare: do both methods flag the same anomalies, or do they disagree? What does this tell you?

Task 6 — Product Demand Segmentation using Clustering

- Aggregate data at the **product sub-category level** with features like:
    - Total sales volume
    - Sales growth rate (year-over-year)
    - Sales volatility (standard deviation of monthly sales)
    - Average order value
- Apply **K-Means Clustering** to segment products into demand groups
- Use the **Elbow Method** to find the optimal number of clusters
- Label each cluster meaningfully, for example:
    - High Volume, Stable Demand
    - Low Volume, High Volatility
    - Growing Demand
    - Declining Demand
- Plot clusters using a 2D scatter plot (use PCA to reduce to 2 dimensions if needed)
- Write: what stocking strategy would you recommend for each cluster?

Task 7 — Deployment: Interactive Dashboard using Streamlit
Build a working Streamlit web app with the following features:

Page 1 — Sales Overview Dashboard

- Total sales by year (bar chart)
- Monthly sales trend line chart
- Sales by region and category (interactive filters)

Page 2 — Forecast Explorer

- Dropdown to select: Category or Region
- Date range slider to select forecast horizon (1, 2, or 3 months ahead)
- Display the forecast output from your best model for the selected inputs
- Show MAE and RMSE of the model below the chart

Page 3 — Anomaly Report

- Display the anomaly chart from Task 5
- List detected anomaly dates in a table with their sales values

Page 4 — Product Demand Segments

- Display the cluster chart from Task 6
- Show which sub-categories belong to which demand cluster in a table

Deploy the app on **Streamlit Community Cloud** (free) and submit the live link.

Task 8 — Executive Business Report

Write a structured 2-page business report (in summary.pdf or summary.docx) as if you are presenting to the **Head of Supply Chain and the CFO** of a retail company. The report must include:

- A one-paragraph executive summary (no jargon)
- Key findings from EDA and forecasting
- The 3-month sales forecast with confidence ranges in plain language
- Top 3 anomalies detected and their likely cause
- Product demand segmentation findings and recommended stocking strategy per segment
- 3 concrete business recommendations with data backing each one
- One risk/limitation of this system that the business should be aware of

This report must be readable by someone who has never opened Python.

🛠️ Tools & Libraries to Use

Tool	Purpose
Python 3.x	Main programming language
Jupyter Notebook / Google Colab	To write and run your code

Pandas & NumPy	Data loading, cleaning, feature engineering
Statsmodels	SARIMA model and time series decomposition
Prophet	Facebook's production-grade forecasting library
XGBoost	ML-based time series forecasting
Scikit-learn	Isolation Forest, K-Means, PCA, evaluation metrics
Matplotlib / Seaborn / Plotly	Static and interactive charts
Streamlit	Interactive dashboard deployment
Git & GitHub	Version control and code submission

📁 **What to Submit**

Create a folder named: **SalesForecasting_[YourName]**
Inside the folder, include:
✅ analysis.ipynb — Complete Jupyter Notebook with all 8 tasks, with markdown explanations throughout
✅ train.csv — The Superstore sales dataset used
✅ app.py — Your Streamlit dashboard code as a separate Python file
✅ requirements.txt — All Python libraries used (so the app can be re-deployed by anyone)
✅ summary.pdf or summary.docx — 2-page executive business report
✅ charts/ — All chart images saved as .png

Submit the entire folder as a ZIP file + paste your Google Colab link + paste your live Streamlit app link
Via: https://docs.google.com/forms/d/e/1FAIpQLSf7t2YhRzjpQi0CL4TtG6gjqGWZ168Rk047SmuPTAXIUB7Flw/viewform?usp=dialog

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📊 **Evaluation Criteria**

Criteria	Weightage
Time Series Analysis & Decomposition	10%
Correct implementation of all 3 forecasting models	20%
Model comparison with correct metrics & justified recommendation	15%
Anomaly Detection (both methods)	10%
Product Segmentation & Clustering quality	10%

Streamlit Dashboard — functionality & usability	15%
Executive Business Report — clarity & business relevance	10%
Code quality, comments, notebook structure	5%
GitHub repository & requirements.txt	5%

Total: 100%

⚠️ **Important Note for the Intern:**
This project is intentionally designed to be uncomfortable. You will encounter concepts you have not seen before — SARIMA parameters, Prophet's data format, converting time series into supervised ML features, PCA for visualization. That discomfort is the point. A real data science role at any company will hand you a problem you have never solved before on your first week. How you handle not knowing something — how you research, attempt, fail, adjust, and document — is what this final project is actually evaluating.

A working project with honest documentation of what you could not fully solve is worth far more than a perfect-looking notebook that was copied without understanding.